Software Engineering & Testing

Sprint (Lab 4)

Team name: “`QWERTY`”

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https://github.com/DanyilT/tu863-wdss-qwerty-social-network

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Annotated Use Case Specifications

Use Case 1: User Registration

- **Actors:** New **User** *(entity)*
- **Description:** Allows a new **user** to create an **account***(entity attribute)*.
- **Preconditions:** The **user** has access to the sign-up page.
- **Flow:**
    1. **User** enters **username**, **email**, and **password** *(entity attributes)*.
    2. System **validates** inputs.
    3. System **stores user information** securely.

Use Case 2: Post Creation

- **Actors:** Registered **User**
- **Description:** Allows **users** to **create** a **post** *(entity)* with text, images, or videos.
- **Preconditions: User** is logged in.
- **Flow:**
    1. **User** clicks the "Create Post" button.
    2. **User** enters **post content** and uploads optional **media** *(entity attributes)*.
    3. System validates input and uploads content.
    4. System **stores** the **post** in the database.
    5. The **post** appears in the **user's feed** .

Use Case 3: Commenting (Additional use case)

- **Actors:** Registered **User**
- **Description:** Allows **users** to **comment** *(entity)* on **posts**.
- **Preconditions: User** is logged in and viewing a **post**.
- **Flow:**
    1. **User** writes **comment text** *(entity attribute)* in the comment field.
    2. **User** submits the **comment**.
    3. System **stores** the **comment** in the database.
    4. **Comment** appears under the **post** *(relationship)*.

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Draft Entity Relationship Diagram

The following entities have been identified from the Use Case Specifications:

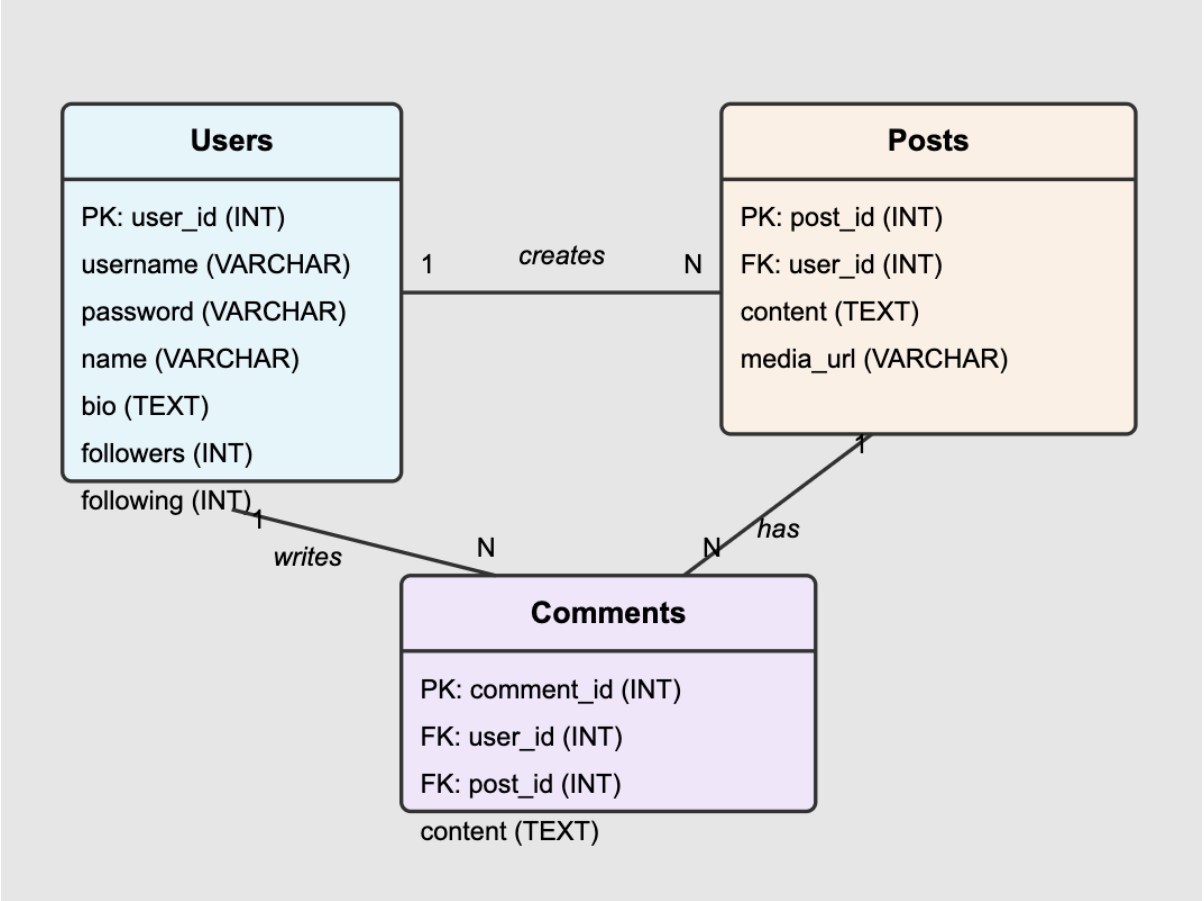
1. **Users** - Identified from "User" in all use cases
2. **Posts** - Identified from "post" in Use Case 2
3. **Comments** - Identified from additional use case for commenting

Entity Attributes Extracted:

- **Users**: user_id, username, email, password, name, bio, followers, following
- **Posts**: post_id, user_id (FK), content, media_url
- **Comments**: comment_id, user_id (FK), post_id (FK), content

Entity Relationships:

- Users (1) to Posts (N): One user can create many posts
- Users (1) to Comments (N): One user can write many comments
- Posts (1) to Comments (N): One post can have many comments



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